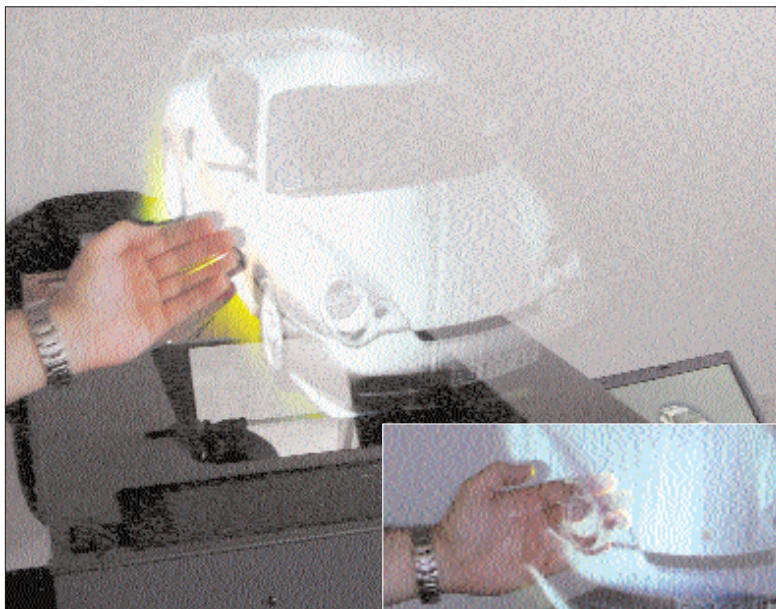




IO2Technology says the Heliodisplay is the world's first interactive video-into-air display. At bottom right is a close-up of the observer's hand 'going into' the image

side. This box sucks in air, 'conditions' it, and then expels the air and uses lasers to project images on to this 'conditioned' air. Sounds vague? It is, and IO2 isn't going to give you any more details about the working of the system just yet. They do, however, inform you that the air is "safe" and say that even if you run the machine for a full day in a sealed room, the air would still be breathable!

Our best guess is that there's ionisation involved, but in the end, we're as clueless as the rest of the world! The fact is that here we have a truly holographic display that works!

So have we attained true 3D Nirvana then? Well, not quite! The problem with IO2's display is that it is still nothing but a computer display. Even with this display, craning your neck to see what's behind that box in a game will not help you see any better. Moving to the side will give you reduced visibility, just like with a computer monitor!

The great thing is that this is a step forward, and it's still a display technology that projects an "emulated 3D image" (as with the 3D in games) into thin air, and allows you to interact with it. You can actually 'touch' your Start button, and pass your hand through the scenery. It's also remarkable that you can connect almost anything with a video-out function to this device!

The Illusion Of Holographic Displays

Laser Magic Productions says on their Web site, "There is no easy way of creating a walking, talking 'holographic projection' in the real world. Real laser-created holograms just don't work that way. There is no current technology to project a real life-sized, moving, talking hologram in a public place that permits a person to walk all the way around and view it from all sides.

The virtual 3D world where you can project anything you want into thin air like The HoloDeck of *Star Trek* fame is still a long way off.

Laser Magic Productions of Playa del Rey, California, has developed a family of 3D projection technologies that accomplish the illusion of Holographic 3D Projection."

"Illusion"? Yes, there are five technologies that Laser Magic Productions mentions, all of which produce the "illusion" of 3D projection. These include the HoloTank, a 3D volumetric display created by projecting scanned laser light or vector video graphics into a liquid medium resulting in 3D holographic-type projections. Moving laser or video images are projected into a liquid chamber (like an aquarium) in which proprietary microscopic particles are suspended. When vector graphics such as words, line art, cartoons and abstract geometric images are projected into the Tank, they are imaged in 3D, resulting in holograms within the Tank.

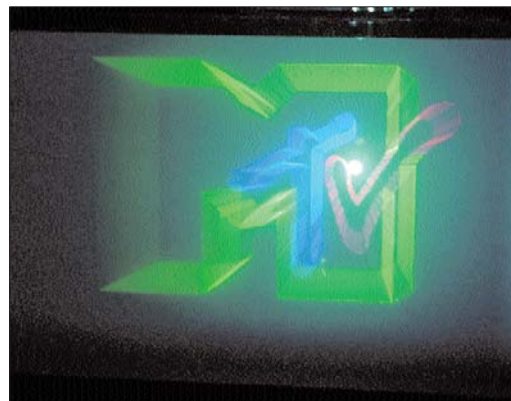
Another innovation by Laser Magic Productions is the LaserCube, a way of displaying laser imagery in 3D space *without* particulates in the air. Images projected into the LaserCube appear to float in space. To give you a flavour for the kind of technology that goes into laser imagery, a LaserCube consists of two elements: the Imaging Cube and the Projector. The Projector is what projects the images, and the images are formed within the Cube.

This Imaging Cube is made up of multiple parallel imaging screens created from a fibre-optic-like material. This material acts as the holographic medium—remember that for a hologram to be formed, a medium is always required!

Now, when laser light is projected into the Cube, the light forms an image on the first of the imaging screens, and then passes through to the next—and so on. The same image appears on each of several parallel planes, creating the holographic illusion of depth and perspective.

What about the Projector? That's where the images come from. How does it work? There are several modes of operation, and in one mode, three lasers are mounted within the Projector. These produce red, yellow and green laser beams. The three beams are directed through a scanning system that creates *abstract* laser images. These have an interesting property: they move and change in response to sound—be it music, voice or even ambient audio.

Now, one can play something so that it acts as the sound source, or, ambients sounds (if any)



The HoloTank can project video images into a liquid medium to produce 3D video images with real depth